

7-Day Learning Plan: VS Code + Python + AI Agents

Goal: Go from total beginner to having a working agent + backend + dashboard skeleton, ready to build a full AI-agent platform.

Pace: ~2-4 hours/day. Don't rush Day 3 or Day 4 if you need more time — they're the foundation everything else depends on.

Day 1 — VS Code + Python Environment Setup

- ☐ Install Python (latest stable version)
- ☐ Install VS Code
- ☐ Install the Python extension (Pylance) in VS Code
- ☐ Learn the interface: file explorer, integrated terminal (`Ctrl+``), command palette (`Ctrl+Shift+P`)
- ☐ Run a script using the green "play" button
- ☐ Create a virtual environment: `python -m venv venv`
- ☐ Understand why virtual environments matter
- ☐ Install GitLens and Prettier extensions (for later)

Day 2 — Python Fundamentals for Automation

- ☐ Variables, loops, functions
- ☐ Lists and dictionaries
- ☐ Working with JSON (`json.loads`, `json.dumps`)
- ☐ `try` / `except` error handling
- ☐ Practice: write small functions, run them, and read tracebacks when they break

Day 3 — Calling an LLM API from Code 🌟 (most important day)

- ☐ Get an Anthropic (or OpenAI) API key
- ☐ Store the key safely in a `.env` file
- ☐ Install `python-dotenv`
- ☐ Write a script that sends a prompt to Claude and prints the response
- ☐ Understand: system prompt, message history, streaming responses

Day 4 — Tool Use / Function Calling (the "agent" part)

- ☐ Understand what makes a script an "agent" (tool/function calling)
- ☐ Write 1-2 simple Python functions (e.g. `get_weather()`, `search_notes()`)
- ☐ Build the loop: model requests a tool → your code runs it → result goes back to the model
- ☐ Test your mini agent end-to-end

Day 5 — Backend Basics with FastAPI

- ☐ Install FastAPI and `uvicorn`

- ☐ Create a `/chat` endpoint that takes a message and returns the agent's reply
- ☐ Test the endpoint using the auto-generated docs page (`/docs`)
- ☐ Confirm your agent works through an HTTP request, not just a script

Day 6 — A Simple Dashboard / Frontend

- ☐ Choose: Streamlit (pure Python, fast) OR a plain HTML/JS page with `fetch()`
- ☐ Build a basic chat-style UI
- ☐ Connect it to your Day 5 FastAPI backend
- ☐ See agent responses appear in the UI

Day 7 — Git, Structure, and Planning the Real Build

- ☐ Learn git basics: init, add, commit, push
- ☐ Use VS Code's Source Control panel (not just terminal)
- ☐ Push your project to GitHub
- ☐ Organize code into folders: `agent/`, `api/`, `frontend/`
- ☐ Research CrewAI, LangGraph, or Anthropic's Agent SDK for your next step (multi-agent orchestration)

After This Week

You'll have a working skeleton: agent → backend → dashboard. This is the exact foundation a full AI-agent platform grows from. The next phase is layering in multi-agent orchestration, persistent storage, auth, and deployment.